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ROLL NO : 50
BATCH : SE B2

Assignment 03: Assignment and practice of ChatGPT and its usage

LAB 3: Precision Prompting for Information Extraction

PROMPT 1 : Rahul Sharma is a software developer. He lives in Pune. His email address is rahul@example.com and his phone number is 9876543210. He has skills in Python, Java, SQL and Machine Learning.

OUTPUT :

Thanks. You've provided Rahul Sharma's personal and professional details. What would you like me to create from them—for example, a resume, professional bio, LinkedIn summary, or profile?

PROMPT 2 : Extract the person's name, occupation, city, email, phone number, and technical skills from the above text. Present the information in a table.

PROMPT 3: Extract the person's name, occupation, city, email, phone number, and technical skills. Return the result only as JSON.

LAB 4: Summarization and Text Transformation

PROMPT : Summarize the above article in 100 words. Include only the main points.

Abstract Artificial Intelligence (AI) is a rapidly developing field of computer science that enables machines to perform tasks that traditionally require human intelligence, including learning, reasoning, pattern recognition, prediction, and decision-making. Recent research demonstrates that AI technologies such as machine learning, deep learning, natural language processing, and generative AI are increasingly being applied in healthcare, education, business, and other sectors. In healthcare, AI has shown potential for supporting disease diagnosis, clinical decision-making, medical training, and personalized learning. Studies also indicate that AI-based educational tools can improve certain learning outcomes and practical skills; however, the available evidence remains limited and requires stronger methodological validation. Despite its benefits, the implementation of AI presents challenges related to data privacy, ethical responsibility, bias, accuracy, transparency, and human-AI collaboration. Therefore, future research should focus on developing reliable, explainable, secure, and ethically responsible AI systems. Overall, AI has significant potential to improve efficiency, innovation, and decision-making across multiple domains, provided that its development and implementation are supported by rigorous research and appropriate human oversight.

OUTPUT :

Artificial Intelligence (AI) enables machines to perform tasks requiring human intelligence, such as learning, reasoning, prediction, and decision-making. AI technologies, including machine learning, deep learning, natural language processing, and generative AI, are increasingly used in healthcare, education, business, and other fields. In healthcare, AI supports diagnosis, clinical decisions, training, and personalized learning. AI-based educational tools may improve learning outcomes, though stronger research is needed. Major challenges include privacy, bias, ethics, accuracy, transparency, and human-AI collaboration. Future AI systems should be reliable, secure, explainable, and ethical. Human oversight is essential for responsible AI implementation.

LAB 5 : ChatGPT in Code Generation and Debugging

PROMPT : Write a Python program to calculate the factorial of a number entered by the user. Include comments explaining the code.

```
# Program to calculate the factorial of a number

# Take a number as input from the user
num = int(input("Enter a number: "))

# Initialize factorial to 1
factorial = 1

# Calculate the factorial using a for loop
for i in range(1, num + 1):
    factorial = factorial * i

# Display the result
print("Factorial of", num, "is", factorial)
```

Example Output:

```
Enter a number: 5
Factorial of 5 is 120
```

PROMPT : Find the error in this Python program, explain why it occurs, and provide the corrected program.

```
num = int(input("Enter a number: "))
factorial = 0 for i in range(1, num + 1):
factorial = factorial + i
print("Factorial of", num, "is", factorial)
```

The error is in the **factorial calculation logic**.

Error

```
factorial = 0
```

and

```
factorial = factorial + i
```

These statements **add** the numbers instead of **multiplying** them. Factorial requires multiplication:

5! = 5 × 4 × 3 × 2 × 1 = 120

Also, the initial value must be **1**, because multiplying by 0 would always give 0.

Corrected Program

```
# Program to calculate factorial

num = int(input("Enter a number: "))

# Factorial starts with 1
factorial = 1

# Multiply numbers from 1 to num
for i in range(1, num + 1):
    factorial = factorial * i

# Display the result
print("Factorial of", num, "is", factorial)
```

Example:

```
Enter a number: 5
Factorial of 5 is 120
```

In short: Change `factorial = 0` to `factorial = 1` and change `factorial + i` to `factorial * i`.

LAB 6 : Domain-Specific Applications

PROMPT : Create 10 multiple-choice questions about Python programming for first-year computer science students. Each question should have four options and one correct answer.

OUTPUT :

10 Multiple-Choice Questions: Python Programming

1. **Which symbol is used to write a comment in Python?**
   A) `//`
   B) `#`
   C) `/*`
   D) `--`
   Correct Answer: B) `#`
2. **Which of the following is used to display output in Python?**
   A) `display()`
   B) `echo()`
   C) `print()`
   D) `output()`
   Correct Answer: C) `print()`
3. **Which of the following is a valid Python variable name?**
   A) `2number`
   B) `my-name`
   C) `my_name`
   D) `class`
   Correct Answer: C) `my_name`
4. **What is the data type of `10` in Python?**
   A) `float`
   B) `string`
   C) `integer`
   D) `boolean`
   Correct Answer: C) `integer`
5. **Which operator is used for exponentiation in Python?**
   A) `^`
   B) `**`
   C) `//`
   D) `%`
   Correct Answer: B) `**`
6. **What will `print(5 + 3)` display?**
   A) `53`
   B) `2`
   C) `8`
   D) `15`
   Correct Answer: C) `8`

7. **Which statement is used to make a decision in Python?**
   A) `for`
   B) `if`
   C) `def`
   D) `import`
   Correct Answer: B) `if`
8. **Which loop is commonly used to iterate over a sequence in Python?**
   A) `repeat`
   B) `foreach`
   C) `for`
   D) `loop`
   Correct Answer: C) `for`
9. **Which function is used to get input from the user in Python?**
   A) `get()`
   B) `input()`
   C) `read()`
   D) `scan()`
   Correct Answer: B) `input()`
10. **Which of the following is a Python list?**
    A) `(1, 2, 3)`
    B) `{1, 2, 3}`
    C) `[1, 2, 3]`
    D) `<1, 2, 3>`
    Correct Answer: C) `[1, 2, 3]`

PROMPT 2: Act as a computer science teacher. Create 10 multiple-choice questions on Python programming for first-year diploma students. Cover variables, data types, operators, conditional statements, loops, functions, and lists. Each question must have four options (A, B, C, D), exactly one correct answer, and a difficulty level of Easy, Medium, or Hard. Provide the answer key at the end.